

Ahmad Saeed Zaidi

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SUMMARY

Full-Stack Engineer with end-to-end ownership across AI, robotics, graphics, and simulation. Builds from first principles, with a strong focus on the mathematics and systems beneath the surface.

EDUCATION

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI)

Topi, Pakistan

Bachelor of Science in Artificial Intelligence

Aug 2023 - May 2027

— **CGPA: 3.79/4.00** — **Rank 1 in Class** — Dean's Honor List (x5) — **Ranked Top 50 Nationally in Mathematics (NMTC)**

— **Relevant Coursework:** Numerical Analysis, Calculus II, Linear Algebra, Probability and Statistics, Differential Equations

— **Certifications:** Deep Learning, Advanced Mathematics, Web Development (Coursera)

EXPERIENCE

Dune AI

Lahore, Pakistan

Software Engineer Intern, Autonomous Robotics

Jun 2025 - Sep 2025

— **Sensor Fusion and State Estimation:** Developed a 200 Hz Extended Kalman Filter (EKF) pipeline in C++ for deployment on **Raspberry Pi 5** and **STM32 Nucleo** microcontrollers, owning the full life cycle from requirements through 660+ hours of autonomous field testing to achieve sub-5 cm positional accuracy.

— **Vision System Deployment:** Reverse-engineered a research-grade C++ vision framework to calibrate a monocular camera system in unstructured field environments.

— **Trajectory Generation:** Implemented the motion planning layer using TinySpline to generate smooth trajectories in 1 to 2 ms, enforcing strict acceleration profiles (under 1 m/s^2).

— **Production Integration:** Collaborated with cross-functional hardware teams to integrate pipelines into a production fleet, reducing sensor calibration time from 4 hours to 30 minutes.

PROJECTS

Autonomous QLoRA LLM Fine-Tuning Pipeline — *Python, PyTorch, CUDA, Unsloth, Hugging Face, ChromaDB*

2026

— Built an asynchronous CUDA dataset aggregator via Hugging Face, mining GitHub repositories to collect 2,000 annotated scripts and compress sequence lengths by 75 percent (8K+ to 2K tokens).

— Fine-tuned Qwen and Nemotron 9B models using Unsloth 4-bit quantization and stratified batching to fit within a strict 15 GB VRAM budget on a free-tier Kaggle T4 GPU.

— Applied targeted rank-8 LoRA adapters to 10 Mamba, MLP, and Attention layers within the Nemotron architecture, yielding a highly efficient 18M parameter adapter footprint.

— Integrated ChromaDB for Retrieval-Augmented Generation (RAG) and completed rank adaptations in under 3 hours, sustaining 97 percent memory utilization while reducing training infrastructure costs to zero.

ViT-Engine-CUDA — *C++, CUDA, Python, Pybind11, PyTorch, Flash Attention 2*

2026

— Engineered a hardware-accelerated Vision Transformer (ViT-Base-16) backend featuring 12 attention heads and 86M parameters, optimized for the NVIDIA Ampere architecture.

— Wrote specialized routines for Patch Embedding and Layer Normalization with hand-tuned CUDA kernels, reaching 99.5 percent precision on the Imagenette validation subset.

— Integrated Flash Attention 2 with asynchronous memory queues, fused shared-memory tiles, and float4 vectorized accesses, quadrupling memory bandwidth.

— Bridged the C++ backend with Python via Pybind11, exposing zero-copy memory buffers and managing the Global Interpreter Lock (GIL) for native PyTorch interoperability.

— Achieved a 3x increase in data throughput while keeping L2 divergence against native PyTorch under 0.001.

Pleiades: Multi-Modal MLOps Pipeline — *Python, Docker, Neon PostgreSQL, Hugging Face*

2025

— Deployed an ingestion pipeline that extracts image, text, and time series data from 50,000+ daily events.

— Designed a tiered storage architecture using Neon Serverless Postgres to efficiently offload blob assets and achieve a **zero-cost cloud footprint**.

— Architected exploration logic and asynchronous runners to maximize semantic diversity within the dataset while maintaining consistent pipeline stability.

— Enabled real-time video analytics for industrial security through Hugging Face model integrations, cutting processing latency from over 500 ms to under 100 ms.

CUDA C++ and Graphics Optimization Portfolio — *C++, CUDA, OpenGL, GPU Optimization*

2025

— Implemented 10+ core CUDA algorithms (reductions, prefix sums, GEMM) following architectures from *Programming Massively Parallel Processors*, maximizing bandwidth via shared memory and warp-level intrinsics.

— Built a modular 3D graphics engine in C++, OpenGL, and custom GLSL shaders.

— Developed a texture atlas system that batches 8 high-resolution (4K) textures into a single bind, drastically reducing GPU state-change overhead.

— Rendered complex procedural geometry at a stable 60+ FPS, enabling rapid 3D simulation prototyping.

LEADERSHIP AND VOLUNTEERING

Undergraduate Research Group (UROG) — *Tech Lead and DevOps Engineer*

Nov 2025 - Present

Microsoft Learn Student Ambassadors — *Head of Competitive Programming*

Sep 2025 - Present

RoboCup GIKI Team — *Co-Founder, Control Engineer*

Sep 2025 - Present

TECHNICAL SKILLS

Domains: Autonomous Systems, MLOps, Computer Vision, Graphics Engineering, Sensor Fusion, HPC

Languages: C++ (STL, Memory Management), Python, SQL, MATLAB, JavaScript

AI and Robotics: PyTorch, TensorFlow, ROS2, Extended Kalman Filter (EKF), SLAM, Path Planning, Weights and Biases, LoRA

Infrastructure and Systems: Docker, Kubernetes, GitHub Actions, GCP, PostgreSQL, Prefect, Linux (Ubuntu), CMake, OpenGL